Magnetic Fast Steering Mirror for Inter-satellites and Feeder Link Optical Communication

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Abstract

In order to cope with performances of Fine Pointing, and Fast Steering mirror requirements for Space Optical Communication, CEDRAT TECHNOLOGIES (CTEC) is developing magnetic Fast Steering Mirror's (M-FSM) family, based on proprietary MICA™ flexure bearing magnetic actuators mechanism (Moving Iron Controllable Actuator), with embedded eddy current sensors (ECS).
FSM design expected for Optical Inter-Satellite Link (OISL) in large-scale New Space constellations, is highly driven by mandatory cost-efficiency for high quantities recurrent production, while FSM design expected for Feeder-Link communication is driven instead by high power Laser requirements together with high level of redundancy.
In this publication CTEC is presenting both the M-FSM45 and M-FSM-HPL designs, and test results, respectively for inter-satellites and feeder link optical communications.

1 Introduction

Starting from M-FSM62 former design heritage, CTEC has designed first the M-FSM45 dedicated to Giant constellations based on inter-satellites Communication, focusing on mandatory cost efficiency, with high repeatability of performances and reliability, while targeting no defect at customer integration over quantities beyond several thousands of units delivered in very short production time series.
After successful achievement of M-FSM45 Engineering Models design and testing, CTEC is proposing alternate M-FSM-HPL (High Power Laser) design dedicated to Feeder-link applications. The proposed design has been optimized for continuous laser power under vacuum, ranging from 40W to 80W, leading to SiC mirror design with very high reflectivity, and high conductivity. Together with the mirror heat conduction design optimization, a drastic improvement of the FSM heat sinking has been achieved, in order to maintain low equilibrium temperature on mirror surface, together with low thermal gradient, in order to cope with stringent wave front error (WFE) requirement. The FSM size has been increased in order to implement two redundant coils assemblies, each connectable to two separate drive electronics (redundant and nominal).

1.1 M-FSM45 design and tests

The angular stroke amplitude has been increased from +/- 1,5° on M-FSM45 to +/- 2,5° on M-FSM-HPL, together a new concept of eddy currents position sensors, allowing redundancy. Additional end-stops devices have been implemented for the launch vibrations and shock loads, successfully tested to 1,5g²/Hz random vibration level at actuation resonance frequency, and to 1000g SRS shock, avoiding any electrical launch locking device.

Starting from the very first former PYSCHE PAM30 flight project heritage for Deep Space Optical Communication (DSOC), CTEC is developing FSM SiC mirror technology, for both cost-efficient large-scale constellations focusing on mirror cost optimization versus performance, and for Feeder-link high power Laser, focusing on very high reflectivity, and high thermal conductivity.
The environmental tests campaign for Space qualification have been passed, which includes launch vibrations and shock tests, thermal vacuum tests, high frequency accelerated lifetime fatigue tests, and closed loop position control tests, based on advanced control achieved with LQG state feedback controller, in order to reach highest bandwidth possible at high stroke amplitude.

1.1.1 M-FSM45 design description

The M-FSM45 is a magnetic mechanism driving two tilt axes on a large angle requirement. This FSM, which derives from M-FSM62 [4,5] is composed of the following parts:

- A Magnetic circuit in Soft Magnetic Composite material with 4 magnets and a moving part.
- 4 Coils optimized to provide the best induction in the short volume, with potting to dissipate the generated heat.
- An Eddy Current Sensor device with aluminium targets embedded on the moving parts, and 4 sensing heads on a single PCB below.
- A moving part suspended on a flexure bearing ensuring high lifetime performances.
- A mirror fixed on a flexible baseplate limiting the integration deformations.

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The magnetic design relies on forces due to tangential variable magnetic reluctance, which offers higher forces than Lorentz forces [6] and more linear forces than normal variable magnetic reluctance [7].
To ensure the FSM performances, magnetic calculations by FEA have been performed. The magnetic saturation, available torque and parasitic forces were verified.

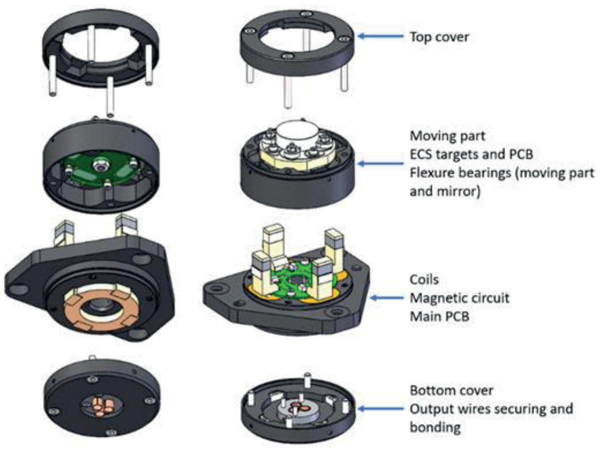

Figure 1: M-FSM cost efficient design concept

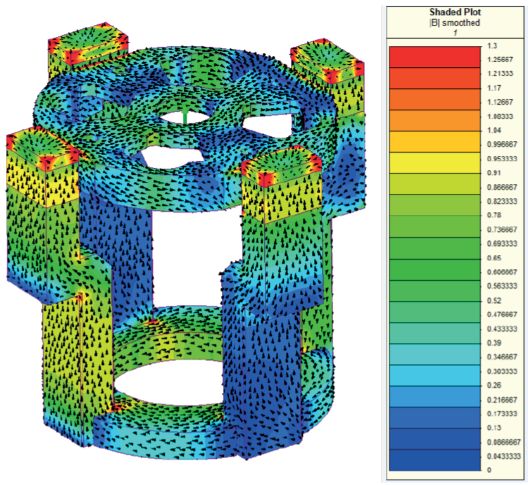

Figure 2: Magnetic circuit finite element modelling

1.1.2 Eddy current position sensors (ECS)

To measure the mirror position and perform closed loop control, an eddy current sensor assembly is embedded in the mechanism. The sensor assembly is eased thanks to the design of a single PCB including the 4 sensing coils, taking advantage of space qualification of PCB-ECS sensors [8]. This solution makes the M-FSM more compact, with an efficient one step assembly. The sensitivity has been optimized for the FSM stroke, making the sensor non-linearity acceptable for the application.

1.1.3 M-FSM45 Manufacturing and Assembly

The mechanism design has been optimized with the objective of reducing the assembly complexity and time, in order to achieve high cost efficiency for very large quantities production. Specific tooling was designed for critical steps such as mirror integration, coils' potting, or moving part assembly into the magnetic circuit.
The magnetic parts were manufactured by industrial molding processes, dedicated to composite magnetic material featuring very low eddy currents. The geometrical tolerances were considered by design to sustain the worst-case air gaps in the torque calculation.

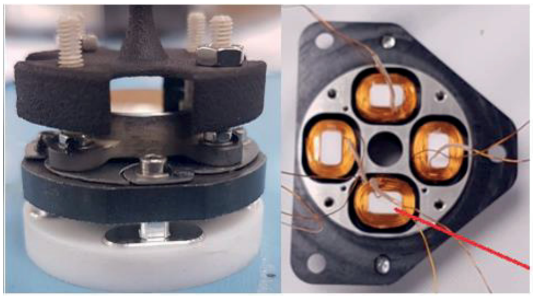

Figure 3: Moving part and coils' assembly

Figure 4: M-FSM45 Engineering Model

The mirrors were manufactured for the engineering models and optical verifications were performed.
The following pictures show the mirrors RWE (reflected wave front error measured with Zygo interferometer at CTEC laboratory).

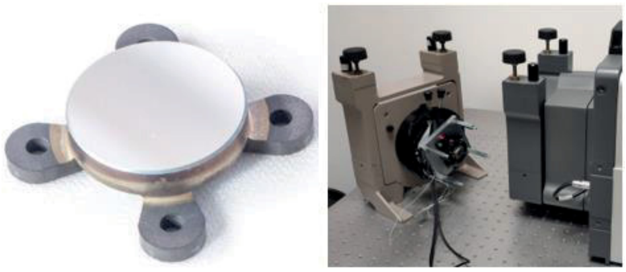

Figure 5: Free state mirror (left) and Mirrors RWE test after integration (right).

After the mechanism assembly, the mirror surface flatness was controlled. The optical verification indicates that the

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mirrors are compliant with important margins in both free state and after integration.

	M-FSM45
Mirror RWE before integration (nm rms)	10
Mirror RWE after integration (nm rms)	12.7

Table 1: mirror optical control results (specification: RWE < 40nm rms)

1.1.4 M-FSM45 tests main results

The M-FSM performances have been measured in open-loop to validate the available stroke, frequency and coils values.
As expected, the first resonance frequency is located around 100Hz. The calculated coils parameters are validated trough the impedance and inductance measurements.

The stroke measured shows that the M-FSM is allowing a maximal stroke slightly lower than +/-1.5° (+/-25.8mrad) for a +/-1A current input. The measurements have been performed thanks to a large angle autocollimator allowing a single angle low frequency acquisition.

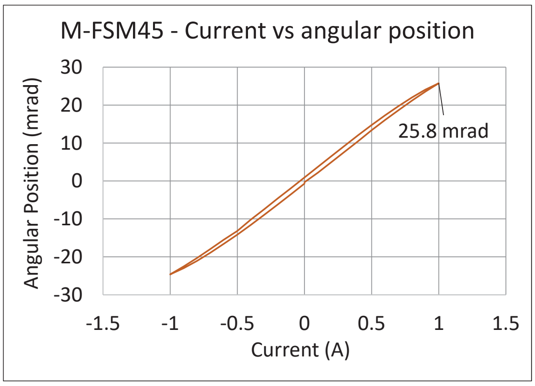

Figure 6: M-FSM45 stroke amplitude test

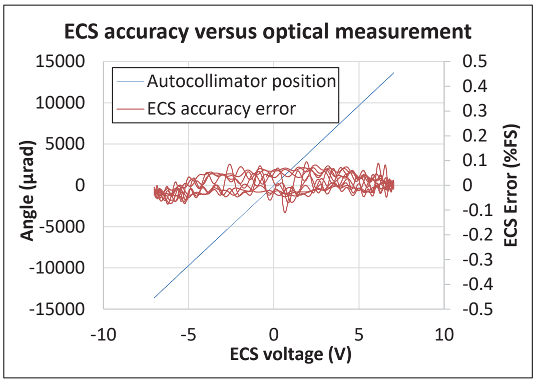

Figure 7: M-FSM45 ECS accuracy test

After complete assembly of the M-FSM45 the ECS position sensors accuracy was measured compared to the optical measure of mirror position with an autocollimator facility. The accuracy error of ECS position sensors was in the range of +/- 0,08% of the +/- 15mrad measurement full scale.

The frequency bandwidth test was achieved at +/-500µrad position amplitude and has shown a -3dB frequency bandwidth up to 320Hz on M-FSM45, which is considered a very good result.

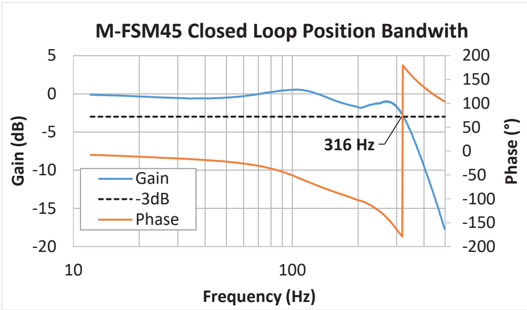

Figure 8: M-FSM45 Closed control bandwidth test

1.2 M-FSM45-HPL design evolution

The M-FSM45-HPL is then derived from the M-FSM45 with some key points added to makes it suitable for High Power Lasers applications.
The functioning principle stays the same with the magnetic circuit in Soft Magnetic composite material to remove the eddy current perturbations, 4 Coils (+ redundant channel) to generate the moving magnetic field, the Eddy current sensor system with a new detection system, a moving part with optimized flexures bearings, and a mirror. As described in Figure 10, the coils will create a magnetic field that will couple with the permanent magnet effect in order to generate a torque on the moving iron.

As for the MFSM45, to hold the mirror on the moving part, a specific flexible element is designed to limit the impact of the screws tightening on the reflected wavefront error, while keeping the mechanism resonances as high as possible.

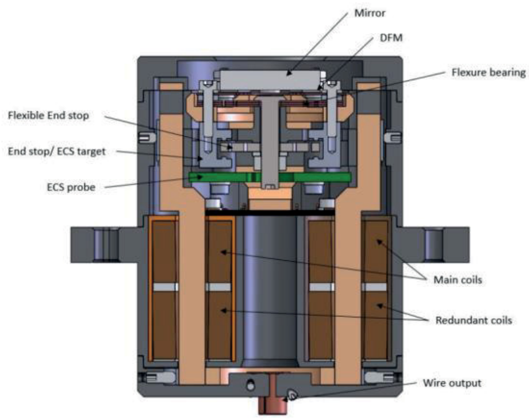

Figure 9: M-FSM45-HPL detailed view

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The -3dB bandwidth expected to be reached by the mechanism is higher than 500Hz.

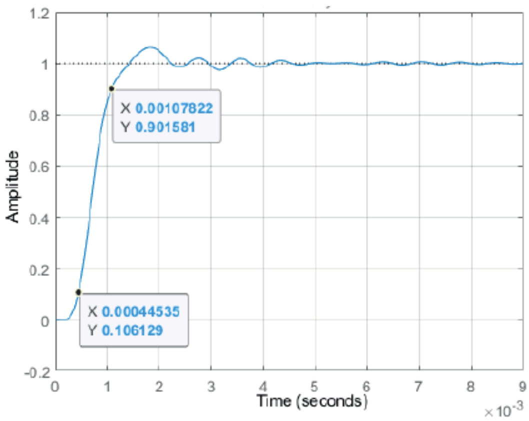

Figure 10: Step and Stay simulation

The mechanism has been reworked in order to optimize the available space inside the 45mm diameter. Thanks to a magnetic redesign, the overall torque is increased, allowing an angle of +/-2.5° while keeping the main resonance frequency higher than 80Hz.

1.2.1 Angle increase

The mechanism has been reworked in order to optimize the available space inside the 45mm diameter. Thanks to a magnetic redesign, the overall torque is increased, allowing an angle of +/-2.5° while keeping the main resonance frequency higher than 80Hz.

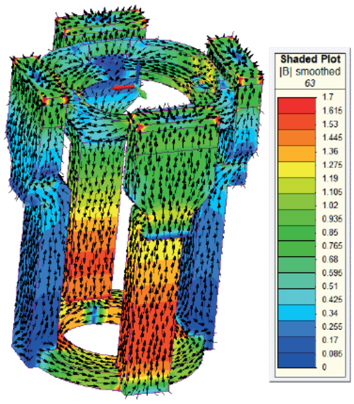

Figure 11: Magnetic field optimisation

These magnetic performances improvements are also visible on the foreseen dissipated power. While the M-FSM45 reaches +/-1.5° for a 4W RMS power at low frequency, the M-FSM45-HPL performs a +/-2.5° angle for a power lower than 0.7W on the redundant channel worst case.

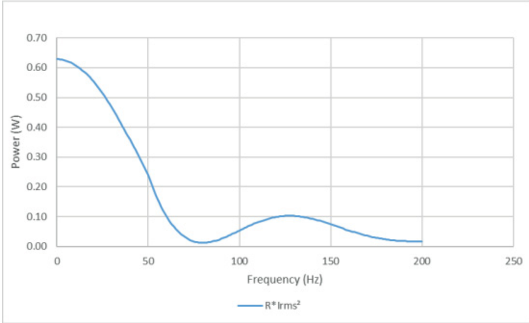

Figure 12: Calculated electrical performances

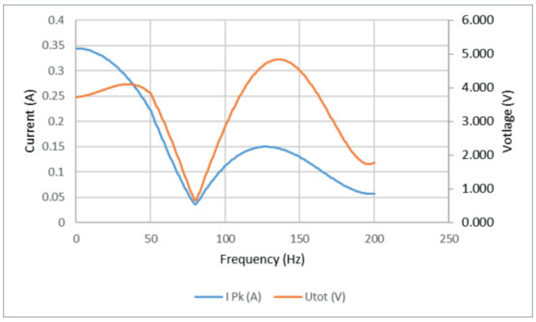

Figure 13: Calculated electrical performances

1.2.2 Thermal optimisation

One of the key performances of the mechanism is to be able to sustain a high power laser up to 80W. The mirror reflectivity is required to be higher than 99.5% to minimize the absorbed power by the mirror. The SiC mirror high thermal conductivity will allow an homogenous thermal distribution inside the mirror in order to reduce the laser effect on the reflected wavefront error. The flexure bearings are designed to dissipate as much heat as possible. Their material is chosen because of its good combination of high thermal conductivity and high yield strength. A stack of a few bearings is preferred instead of a single thicker one to adjust the global tilting stiffness and increase the heat exchange surface.

This design choices allows the mirror to stay under 80°C, which is foreseen to be the maximal temperature for the chosen mirror coating.

1.2.3 Coil redundancy design

The redundancy management for this mechanism is mandatory as it needs to fulfill space flight requirements. The redundant coils system is then added below the main coils channel inside the main coil casing, increasing the length of the mechanism. A cold redundancy solution is chosen here.

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1.2.4 Eddy Current Sensor redundancy design

The ECS system is also reworked to include the redundancy functionality. To improve the mechanism linearity performance a differential mode on a full axis is performed by software after the linearization of each probe individually. As each eddy current probe is independent with a single frequency detection strategy, the redundancy is ensured in case of a failure detection.
The same mechanical design is used, with the probes on a single PCB to ease the integration.

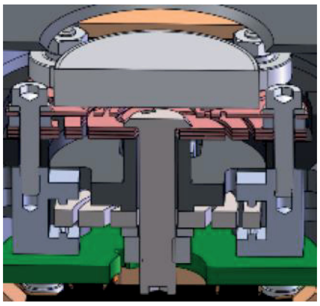

Figure 14: Eddy current sensors integration (PCB in green) implemented on end stops

1.2.5 End stops design

The M-FSM45-HPL is a mechanism with a moving part suspended on its flexure bearings. The high vibration level required for space launch environment is a difficult part of the design. Without additional elements, several notches would be required on the specified levels. The end-stop system designed for the mechanism (Figure 14) allows to sustain the full level of 1,5g²/Hz random vibration at actuation resonance frequency, and up to 1000g SRS shock.
A test campaign has been conducted on a mechanical breadboard with a representative moving part (Figure 15).

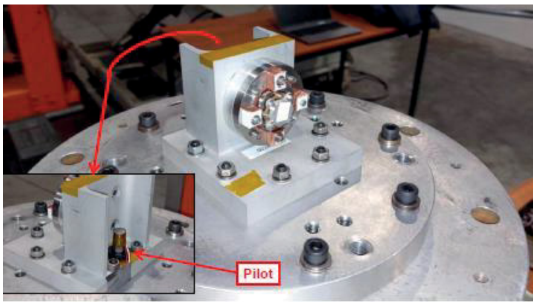

Figure 15: Random vibration test set up

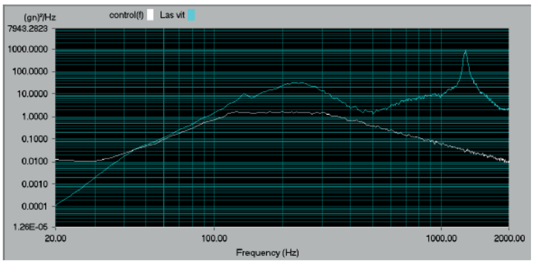

Figure 16: Random vibration PSD test results

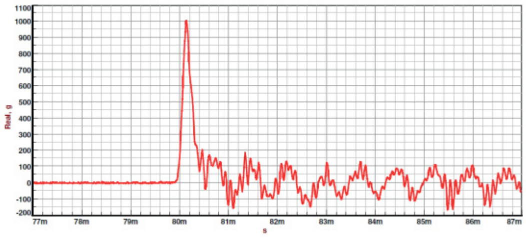

Figure 17: Shock test results

The health of the system is checked before and after the test campaign to validate the good behaviour of the moving part and the flexure bearings. A particle pollution tests has been conducted under the CNES funded COOP program, leading to the conclusion that there is no risk of contamination for the mirror embedded in the mechanism.

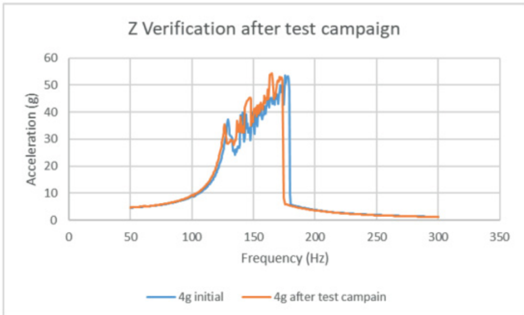

Figure 18: Health check sweeps

1.2.6 M-FSM45-HPL Manufacturing and Assembly

The M-FSM45-HPL tooling have also been reworked to ease the global integration. Specific parts has been designed to allow a mobile part positioning with a high precision and good repeatability. The coils potting is also performed with a specific housing in order to avoid any leakage on critical parts.

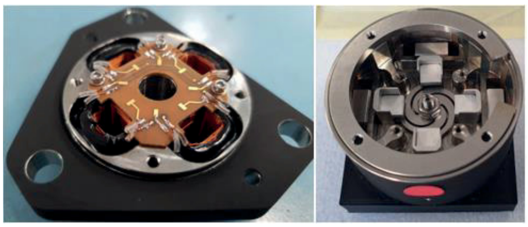

Figure 19: Assembly pictures

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At the time of publication only preliminary tests have been achieved, and complete test campaign results are foreseen by end of 2024. The following pictures shows the tested actuation transfer function versus frequency (fig.20), highlighting the actuation resonance at 80Hz as expected by design, and the +/-45mrad maximum stroke amplitude versus current (fig.21). One can see in fig. 21 the maximum angular range tested up to +/- 47mrad with +/- 0,37A current, measured on autocollimator (fig.22), which illustrates the hysteresis of large stroke magnetic mechanisms. This hysteresis is suppressed by the embedded Eddy Current Sensors when used in position feedback closed loop control.

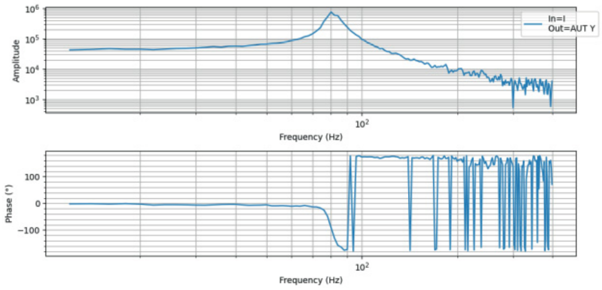

Figure 20: Current to stroke amplitude transfer function

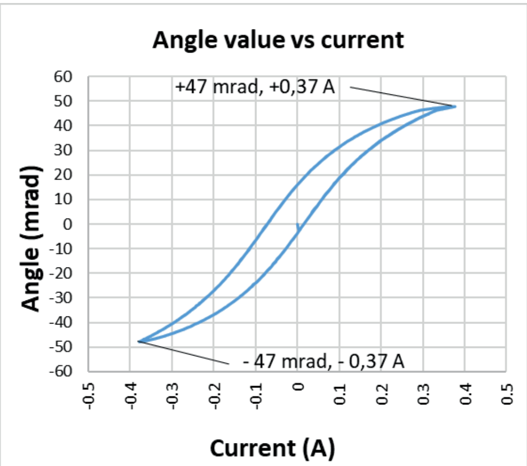

Figure 21: Angular position versus current (open loop)

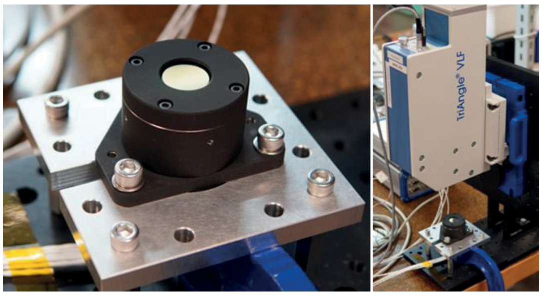

Figure 22: M-FSM45-HPL on autocollimator bench

1.3 Performances summary

As a relevant summary the following table presents the difference of performance of both M-FSM versions.

	Tested on M-FSM45	Expected on M-FSM45-HPL
Mirror aperture	15mm	16.2mm
Housing Diamer	45mm	45mm
Height	35mm	58mm
Mass	195g	370g
Stroke amplitude	+/- 15mrad	+/- 45mrad
Resonance freq.	100Hz	80Hz
Rated voltage	12V	<7V
Rated current	0,5A	<0,4A

Table 2: Performances summary table

1.4 Acknowledgement and conclusion

These performances will be validated through a qualification test campaign that will be performed during 2024. All the key parameters will be checked through functional, environmental (thermal, vacuum) and lifetime tests. All this work was made possible through CO-OP funding via France Relance, with Airbus Defence & Space leading the project in collaboration with CNES.

2 Literature

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